RevMetrix Proposal

Matt Brown, Josh Byers, Charles Carroll, Zach Cox, James Devine, Joseph Downey, Kyle Frankhauser, Emmet Larson, Carson Mack, Gabe Manero, Jakeb Nielsen, Andrew Olvera, Gavin Wentz, Hunter Wolfe, Brandon Woodward

Presentation Overview

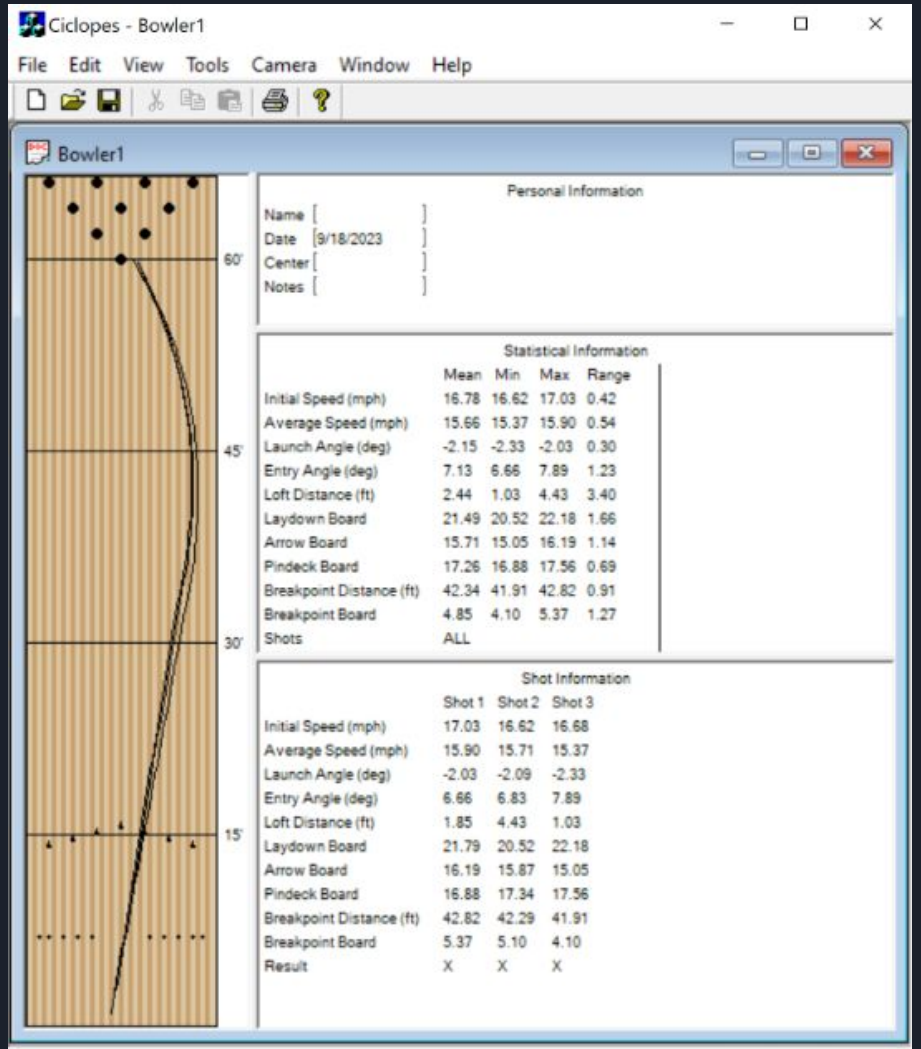
- Background
  - RevMetrix, SmartDot, Ciclopes, Ball Spinner, Mobile App
- Current State of Project
- Proposal
- Technological Challenges
- Conclusion

RevMetrix Background

- Creation of Professor Hake and Dr. Babcock
- Combination of Ciclopes and SmartDot Module
- Both compliment each other with supplemental data
- Application:
  - Bowlers
  - Researchers

What is Ciclopes

- Dr. Babcock's contribution
- External bowling analysis



Personal Information

Name	[ ]
Date	[9/18/2023]
Center	[ ]
Notes	[ ]

Statistical Information

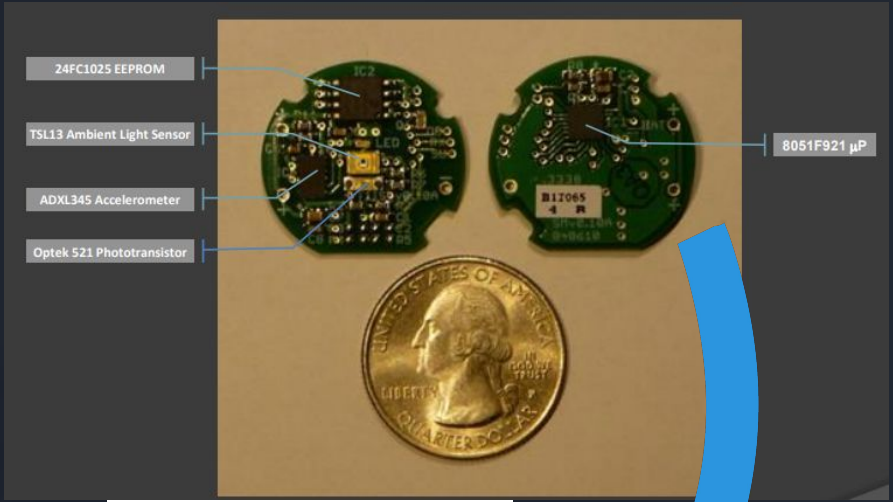
	Mean	Min	Max	Range
Initial Speed (mph)	16.78	16.62	17.03	0.42
Average Speed (mph)	15.66	15.37	15.90	0.54
Launch Angle (deg)	-2.15	-2.33	-2.03	0.30
Entry Angle (deg)	7.13	6.66	7.89	1.23
Loft Distance (ft)	2.44	1.03	4.43	3.40
Laydown Board	21.49	20.52	22.18	1.66
Arrow Board	15.71	15.05	16.19	1.14
Pindeck Board	17.26	16.88	17.56	0.69
Breakpoint Distance (ft)	42.34	41.91	42.82	0.91
Breakpoint Board	4.85	4.10	5.37	1.27
Shots	ALL			

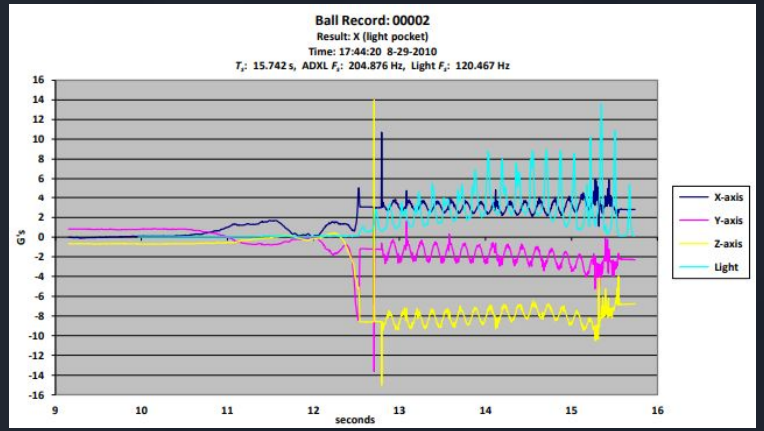
Shot Information

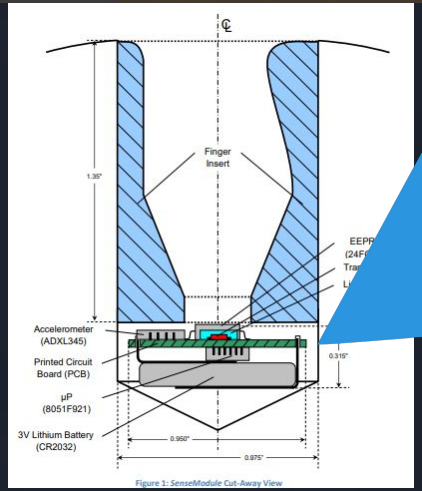
	Shot 1	Shot 2	Shot 3
Initial Speed (mph)	17.03	16.62	16.68
Average Speed (mph)	15.90	15.71	15.37
Launch Angle (deg)	-2.03	-2.09	-2.33
Entry Angle (deg)	6.66	6.83	7.89
Loft Distance (ft)	1.85	4.43	1.03
Laydown Board	21.79	20.52	22.18
Arrow Board	16.19	15.87	15.05
Pindeck Board	16.88	17.34	17.56
Breakpoint Distance (ft)	42.82	42.29	41.91
Breakpoint Board	5.37	5.10	4.10
Result	X	X	X

What is SmartDot

- Professor Hake's Brain-Child
- Light Sensor + Accelerometer

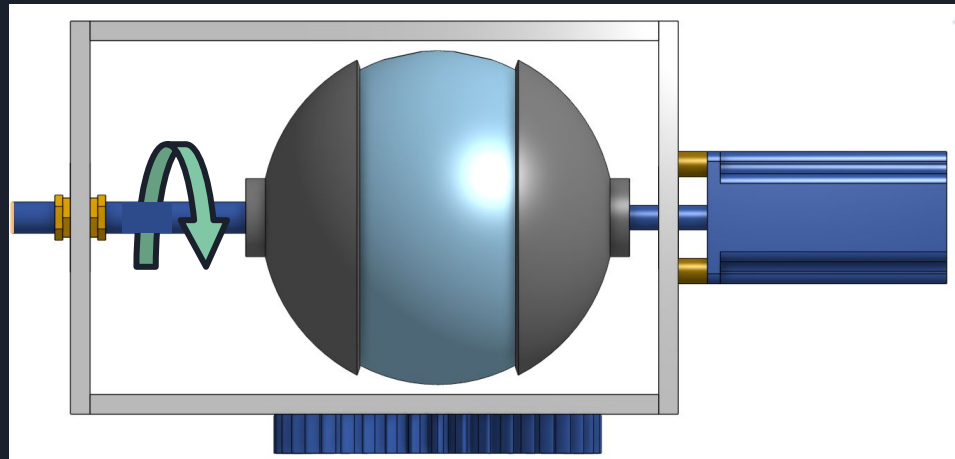






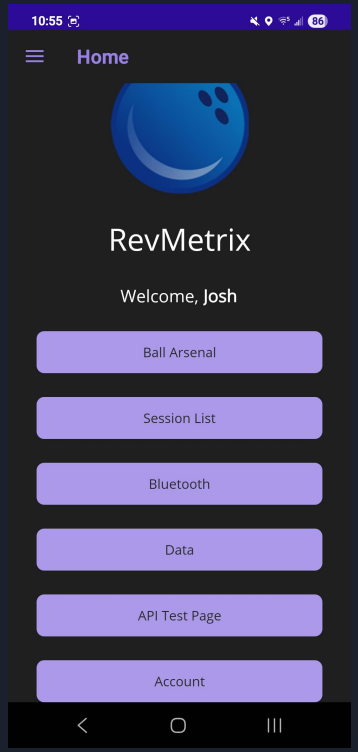
What is the Ball Spinner?

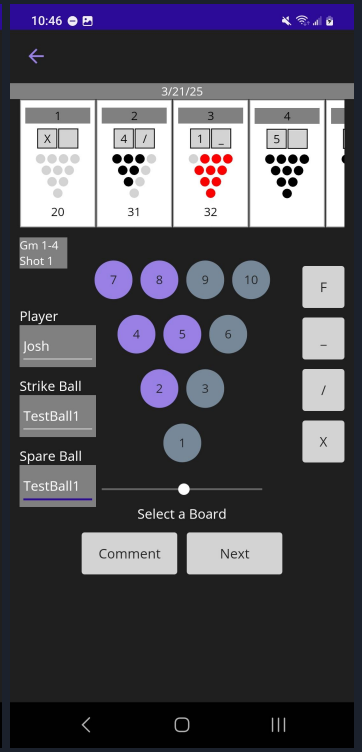
- Device that can emulate a throw down a lane
- Controls movement across 3 orthogonal axes of spin
- External sensors generate data



What is the Mobile App

- Tracking and storing user bowling data
- Viewable bowling stats
- Suggests personalized bowling tips
- Main Connecting to Ciclopes, Watch app, and Smart Dot.



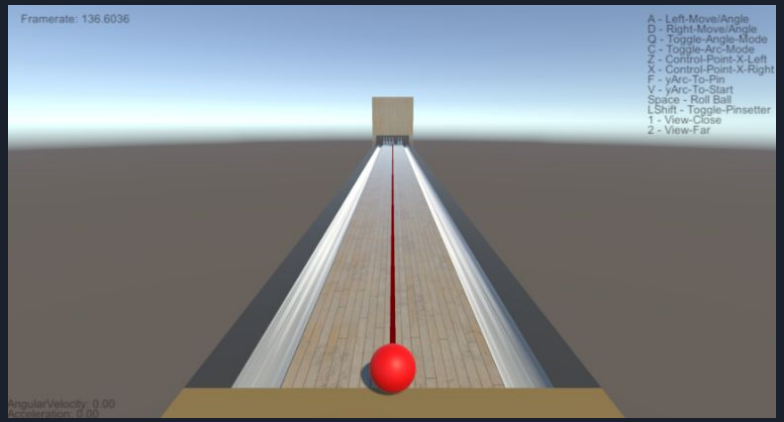


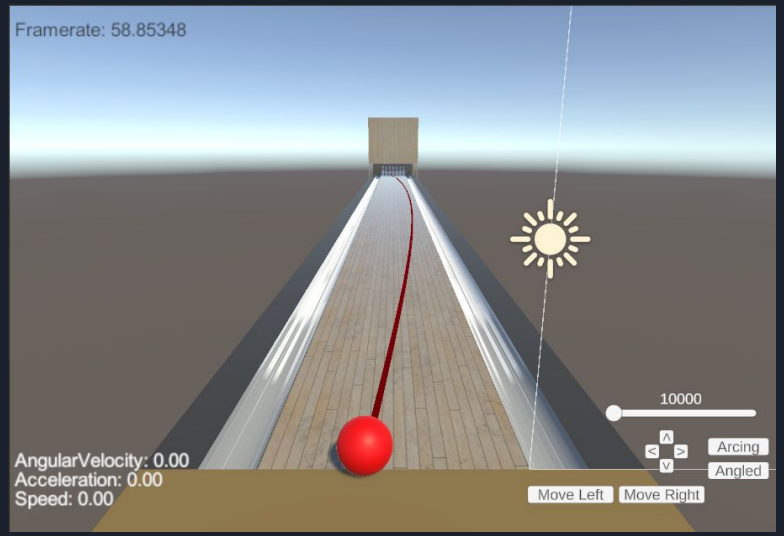
Questions?

Current State of RevMetrix

Unity Simulation

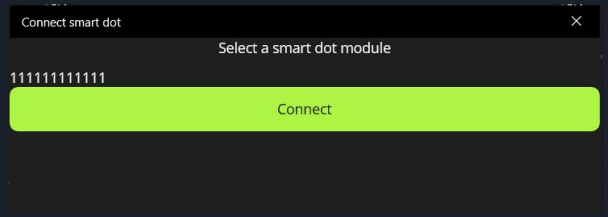
- Barebones bowling functionality
- Basic curve calculation
- Low quality models

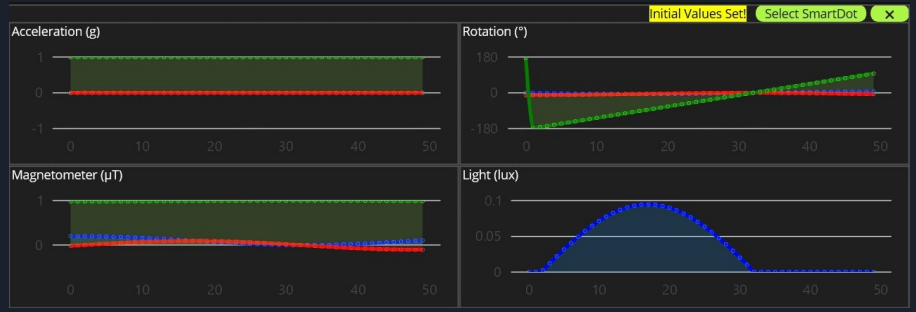


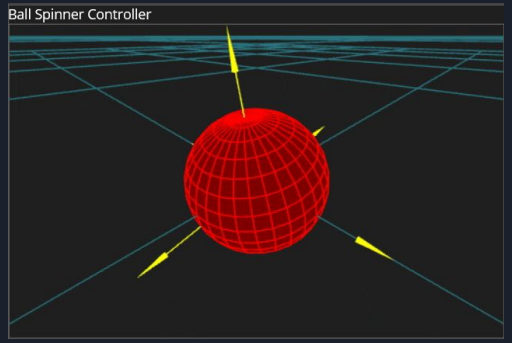


Application Overview

- User Logs in
- User Connects to SmartDot
- Application Spins Motors
- Ball Spinner Controller sends 9Dof Data
- User can save/load previous shot data



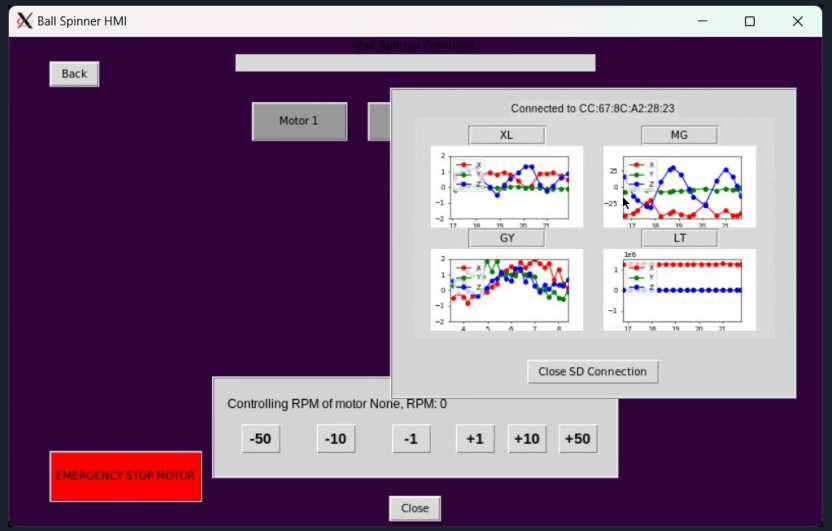




Backend

- Data models for handling sensor data
- Client-side and Server-side API methods
- Local shot feature, saved in CSV files
  - Replay feature
  - Not a RDBMS system
- Testing structure for application and cloud
- Digitalocean and Docker
- CI/CD used to deploy code

Ball Spinner Controller



- Raspberry Pi 4 & PCB
- 2 Motors
- Limited HMI
- BSA Server Connection Issues
- Overscaled hardware
- PCB Supports Expansion for 3 motors

Protocol

Managed By Ball Spinner Controller Team

Protocol General Structure

Msg Type (1 Byte)	Msg Size (2 Bytes)	Msg Data (X Bytes)

Msg Type: Ver.#{XX} Msg.#{XX XXXX}

- Following Messages
  - Initiate Handshake between APP and BSC

APP_INIT

Msg Type (0x01)	Msg Size (0x0001)	Random Start Byte (1 Byte)

  - Send SmartDot MAC Address to Connect To

SMARTDOT_SCAN

Msg Type (0x06)	Msg Size (2 Bytes)	BLE MAC Address (6 Bytes)	Name in ASCII (Up to 255 Bytes)

Protocol

Managed By Ball Spinner Controller Team

- Following Messages
  - Send Motor Instructions

A_B_ABBREVIATED_MOTOR_INSTRUCTIONS

Msg Type (0x0C)	Msg Size (0x0003)	Prime Motor Voltage (1 Byte)	Sec Motor 1 Voltage (1 Byte)	Sec Motor 2 Votlage (1 Byte)

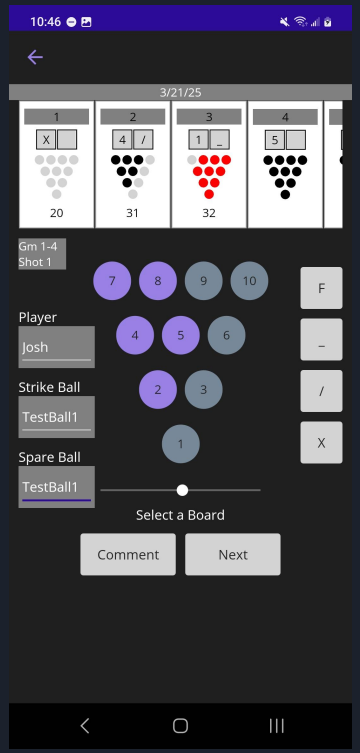
  - Send Sensor Data

B_A_SD_SENSOR_DATA

Msg Type (0x0A)	Msg Size (0x0013)	Sensor Type (1 Byte)	Sample Count (3 Byte)	Time Stamp (4 Byte)	"X-axis" data (4 Bytes)	"Y-axis" data (4 Bytes)	"Z-axis" data (4 Bytes)

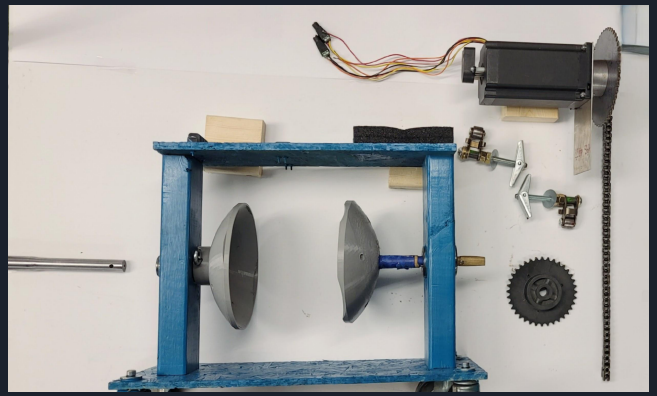
Mobile Application

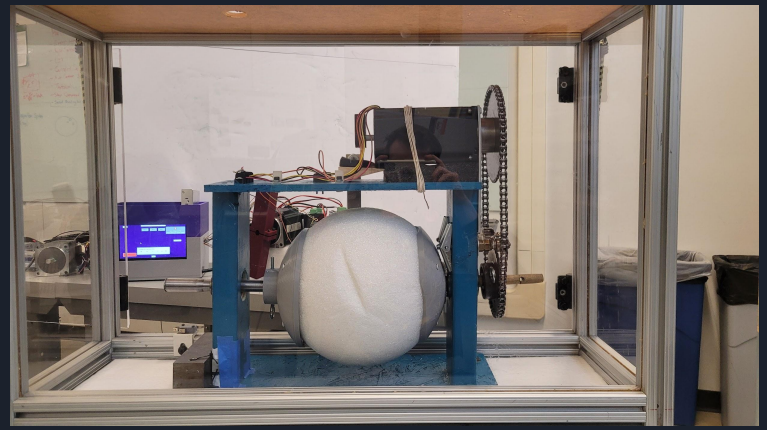
- Create accounts
- Play through full games of bowling
- Display frame
- Allows users to store shot data
- Create events and establishments
- Proof of concept cloud connection



Physical System

- Spin a foam ball
- Stepper motor with chain drive
- Wooden frame
- Housed with plexiglass protective frame





Wiki

https://docs.revmetrix.io/

2024 Capstone

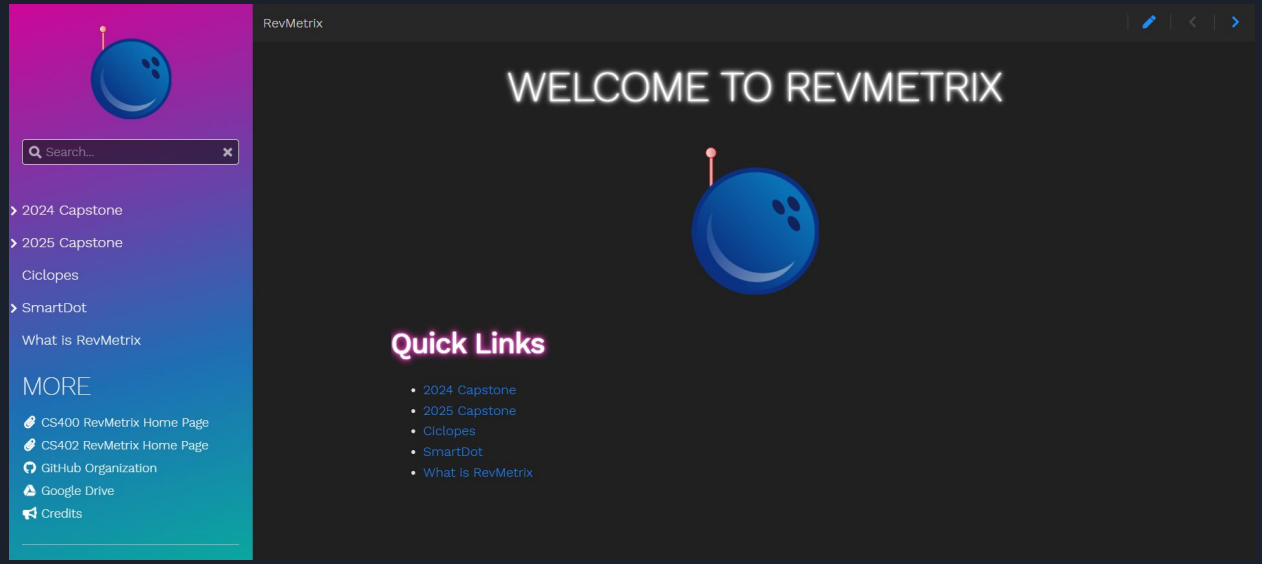
2025 Capstone

- Project assignments
- 2025 Revmetrix
- Technologies used
- Project Milestones
- Set up

Ciclopes

Smartdot

What is Revmetrix



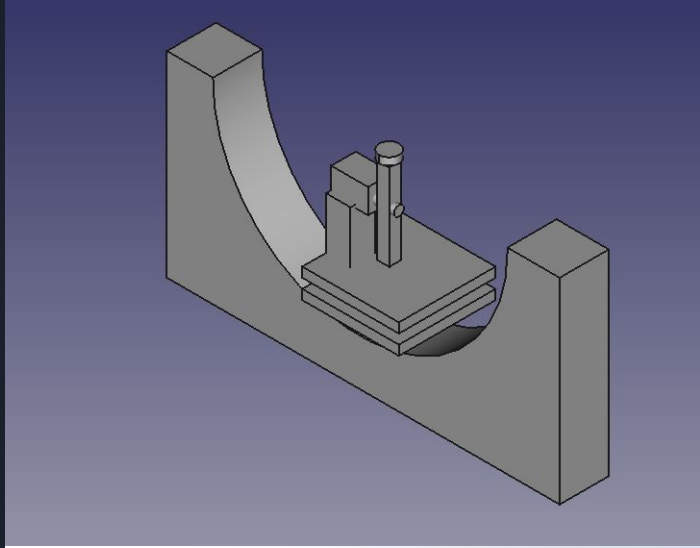
Questions?

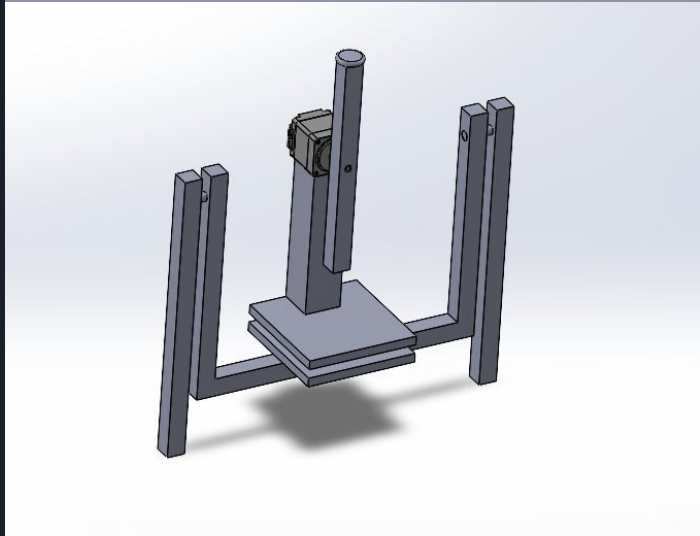
RevMetrix Proposal

Physical Team

James Devine, Joseph Downey, Kyle Fankhauser, Gavin Wentz

- Redesign 1st degree of motion to be more compact
- Design 2nd degree of motion
- Design 3rd degree of motion
- Create CAD assembly of all 3 Degrees of motion
- Prototype 3 degrees separately





Team Pi (Ball Spinner Controller)

Matt Brown, Carson Mack, Gabe Manero, Hunter Wolfe, Brandon Woodward

- Control all 3 motors in the ball spinner
  - Single Shot Mode → emulate bowling shots (2–4s duration).
  - Continuous Testing Mode → real-time control of all motors.
- Replace Tkinter with PyQt
- Integrate Ball Spinner Application (BSA) functionality → streamline into one platform.
  - Graphing data
  - Review of live data and DSP
  - Send data to cloud
- Add a larger touch screen (15.6in or BIGGER)
- Redesign housing for controller
- Upgrade to a faster Pi 5 to accommodate data analysis

BackEnd

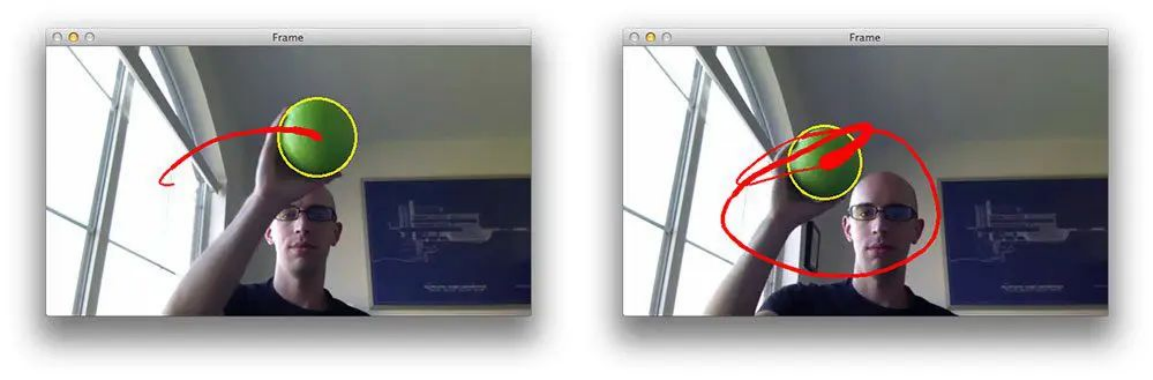
Josh Byers, Zach Cox, Emmet Larson,

- Rework Entire Database with Last Year Scheme
- Testing structure for Phone app, Watch, and Ciclopes
  - Improve test coverage and automation
- Support for other teams

Ciclopes Team

Zach Cox, Carson Mack, Andrew Olvera

- Isaac Sim Synthetic Data Engine
- Training and Evaluation Pipeline
- Robust Util Functions for Homography Math
- Sim Only Implementation
- Mock Microservice with FastAPI using local machine with GPU



Mobile App Team

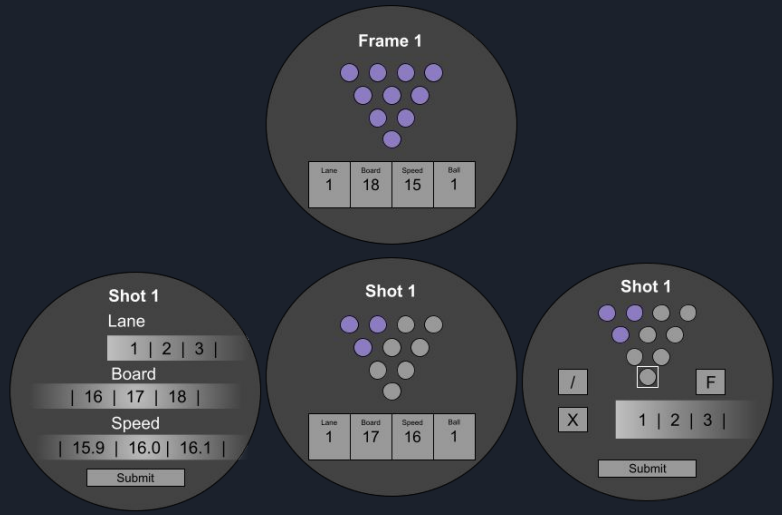
Josh Byers, Charles Carroll, Zach Cox, Emmet Larson, Jakeb Nielsen, Andrew Olvera

- Improved game functionality
- Bluetooth Connectivity
  - Smartwatch
  - Smart Dot
- Full cloud connectivity
- Custom Query Engine

Smart Watch Team

Josh Byers, Charles Carroll, Zach Cox, Emmet Larson, Jakeb Nielsen

- Have functioning prototype that reflects phone app on watch
- Data is stored/retrieved locally
- Basic bluetooth connection established with phone
- Have single piece of data communicating with phone via bluetooth



Wiki

Everyone

- Update color scheme
- Organize content better
- Update as progress is made

Questions?

Technologies Used

Application

- .NET, .NET MAUI
- Three.JS
- Livecharts2
- xUnit

Cloud

- .NET
- xUnit
- Docker
- Digital Ocean
- Github Actions

Ciclopes

- PyTorch (lightning)
- IsaacSim
- Docker

Ball Spinner Controller

- Raspberry Pi
- KiCad
- Tkinter -> PyQt

Ball Spinner

- SolidWorks
- Onshape

Cellular

- .NET MAUI
- SQLite

Smart Watch

- Kotlin

Technical Challenges

Physical Team

- Reduce size
- Ventilation/Heat Dissipation
- Balance
- Safety features
- Implementation of 3 DOF into single assembly
- Stronger material without a large increase in weight
- Vibration suppression

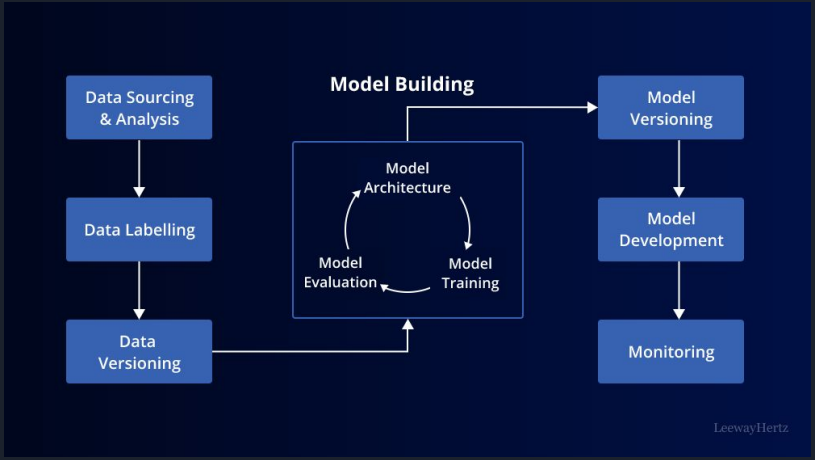
Team Pi (Ball Spinner Controller)

- Having the ball spinner emulate a human shot's physics accurately
- Clean User Interface
- Allowing user to customize shot parameters
- Having functional communications between the Pi and the cloud
- Display data from smart dot module and motor sensors on the display
- Having thorough testing
- Add safety features to the ball spinner (Physical stops, E-stop, Current Limit, ect.)

Backend

- Rebuilding database for Mobile app and Ball Spinner app
- Adding support for future multi-project integration
- Mobile App, Watch, and Ciclopes: no build/test automation yet

Ciclopes Team



- MLops
- Complicated Homography
- Eventual Sim to Real training to bridge gap from sim to real life frames
- Quality of synthetic data
  - Noise in scene (other balls/lanes/aspect ratios) - requires diverse dataset curation
- Accuracy must reach a decided upon threshold
- Due to required accuracy different approaches may be needed if evaluation doesn't meet requirements

Mobile App Team

- .Net Maui
- Porting Java API to C# for smart dot connectivity
- Fully connecting the cloud database to the app
- Communicating with a smart watch

Smart Watch Team

- Android Studio/Kotlin
- Communicating with mobile app (phone) and working in sync with the team to make sure design stays relatively similar
- Configuring on watch itself (UI/screen size, storage, computing power, data transmission)

Wiki

- Teaching new members how to update the wiki
- Documenting issues/ progress
- Finding time to update

Questions?

Many!